

BRIEFINGS

black hat

BLIND TRUST IN THE 6 GHZ BAND

Weaponizing Wi-Fi Automated Frequency Coordination (AFC)

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PennState



About Us



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- Research Assistant, The Pennsylvania State University
- Wireless protocol and systems security, applied cryptography, vulnerability discovery: 5G systems, Wi-Fi AFC systems
- yilud.me



Tianchang Yang

- Research Assistant, The Pennsylvania State University
- Wi-Fi systems, Mobile network security, resiliency, and robustness: 5G, Open RAN, baseband (fuzzing, program analysis, ML)
- tianchang-yang.github.io

Interference Broke a Life-Safety Network

Miami-Dade built a new 6 GHz public-safety microwave system for first responders, engineered for **99.999% reliability**. **It couldn't go into service.**



Interference Broke a Life-Safety Network

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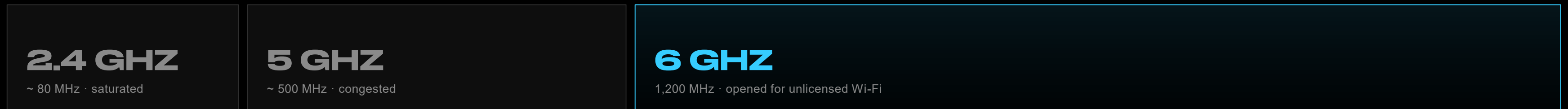
AND THIS WAS AN ACCIDENT

The offenders were unintentional and never hid

What if an attacker induces the same interference on purpose, from a legally-authorized AP device/network transmitting within the rules — then stops and leaves nothing to trace

The Biggest Spectrum Release in Decades

font size



In 2020 the FCC opened **1,200 MHz** at 6 GHz (5.925–7.125 GHz) for commercial use — more contiguous spectrum than everything below it combined, released almost overnight for Wi-Fi 6E and Wi-Fi 7.

The Biggest Spectrum Release in Decades

font size

2.4 GHZ

~ 80 MHz · saturated

5 GHZ

~ 500 MHz · congested

6 GHZ

1,200 MHz · opened for unlicensed Wi-Fi

In 2020 the FCC opened **1,200 MHz** at 6 GHz (5.925–7.125 GHz) for commercial use — more contiguous spectrum than everything below it combined, released almost overnight for Wi-Fi 6E and Wi-Fi 7.

6 GHZ WAS NEVER EMPTY

Fixed Service (FS) microwave links have carried mission-critical traffic across this band for decades. Wi-Fi is the newcomer moving in on top of them.

PUBLIC SAFETY



CONTROL SIGNALS



SCIENCE



MOBILE BACKHAUL



40× Transmit Energy

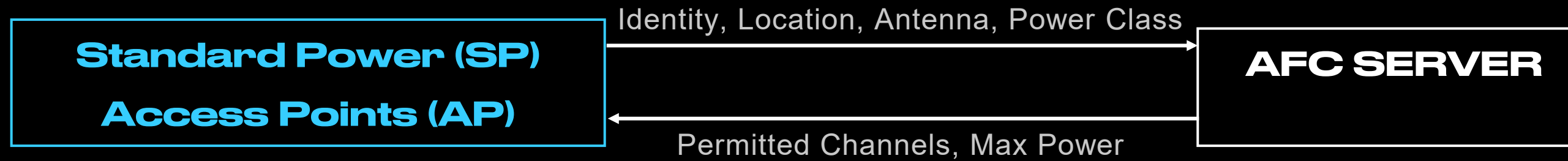
- **Standard Power (SP)** APs transmit far hotter than indoor units: up to **36 dBm EIRP** (≈ 4 W).
- That power buys range — but also the ability to **reach an incumbent's receiver kilometers away**.
- So the FCC forbids an SP AP from transmitting until it **asks permission**.

Low-power Indoor AP	≤ 30 dBm
USRP B210 Jammer	~ 20 dBm
Standard Power Max	36 dBm



Every +3 dB doubles radiated power.
36 vs 20 dBm $\approx$ 40× the energy

AFC: Automated Frequency Coordination



- Simplifies and automates similar systems in other bands (e.g., CBRS in 3.5 GHz bands)
- The AP reports its identity, location, antenna height, and power class to the AFC server.
- The server checks that location against a **database of protected incumbent receivers** and a propagation model, and works out which channels and power levels won't interfere with them.
- AFC returns an **allocation**: channels and the maximum power the AP is allowed to use.
- The allocation is a **lease valid for 24 hours**. When it expires, the AP has to ask again before it can keep transmitting.

AFC Request/Response Format

AP → AFC:
WHO AM I & WHERE I AM

```
AVAILABLESPECTRUMINQUIRYREQUEST
```

```
"location": { "ellipse": {  
  "center": { "latitude": 40.7935,  
    "longitude": -77.8684 },  
  "majorAxis": 41, "minorAxis": 41 } },  
"deviceDescriptor": { "serialNumber": ... }
```

AFC → AP:
THE POWER & FREQUENCY
YOU CAN USE

```
AVAILABLESPECTRUMINQUIRYRESPONSE
```

```
"availableChannelInfo": [{  
  "channelCfi": [1, 5, 9, 13, ...],  
  "maxEirp": [32.4, 32.4, ...] }],  
"availabilityExpireTime":  
  "2025-09-18T18:58:56Z" // +24h
```

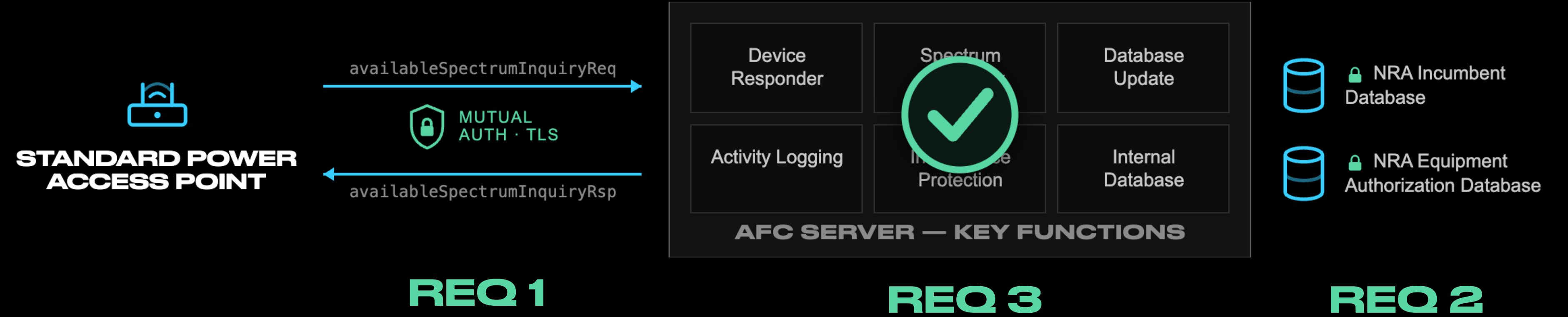
**A SYSTEM WITH
NOTHING TO BREAK**

Simple Protocol & Secure Protection

REQ 1 Mutual authentication between the AP and the AFC server

REQ 2 Integrity of the AFC's internal databases

REQ 3 Accurate interference protection computed on the AFC server



Four Assumptions Hold Up 6 GHz

① HONEST LOCATION

The AP reports its true geographic location.

Assumed.

② HONEST PARAMS

Antenna height and power class match physical reality.

Assumed.

③ UNTAMPERED CODE

Firmware wasn't modified to falsify what it reports.

Crypto ✓

④ AUTHENTIC CHANNEL

The AP↔AFC link preserves authenticity + integrity.

Crypto ✓

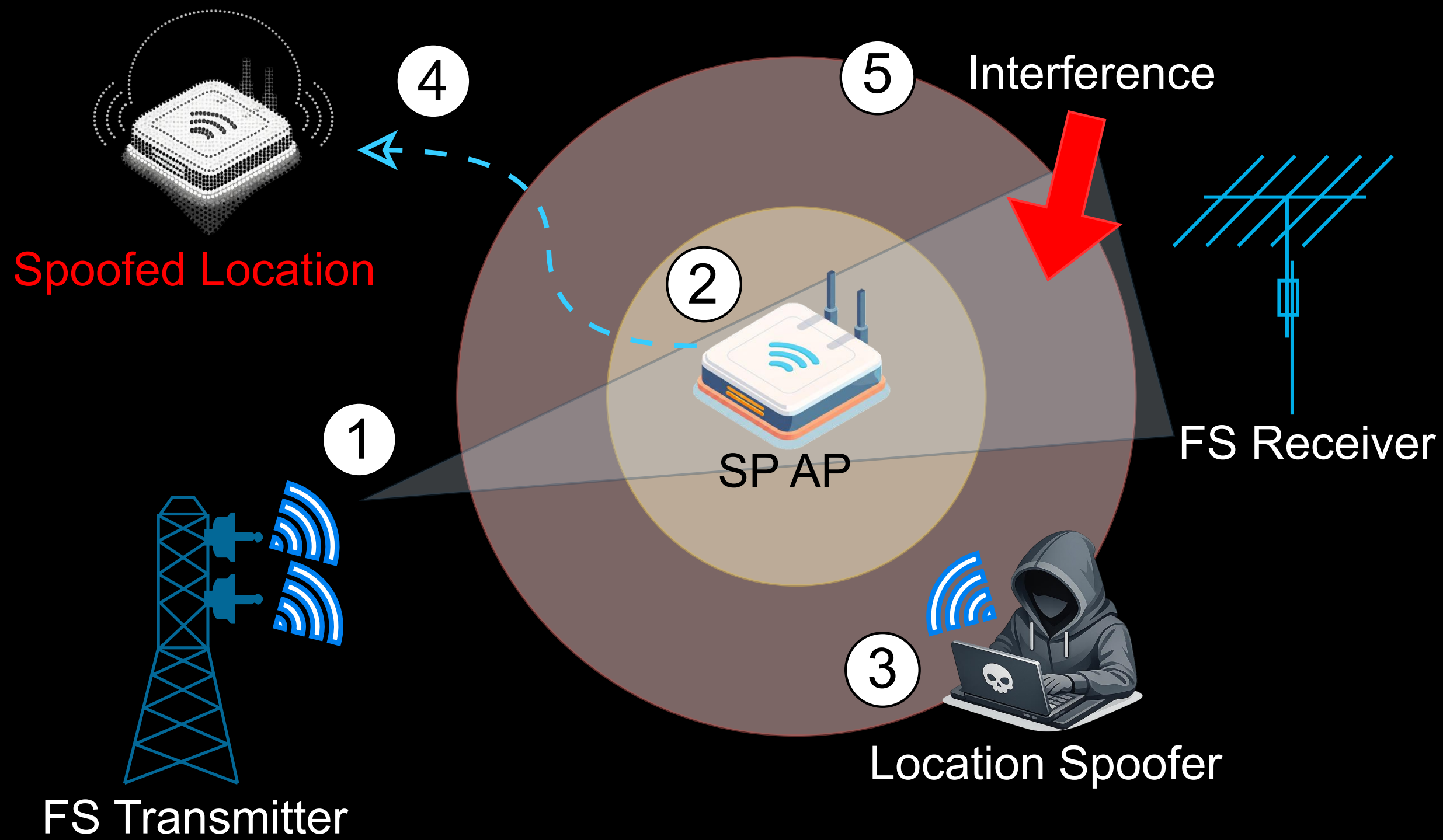
The spec mandates ③ and ④. It spends **nothing** verifying ① and ②.

Threat Model 1

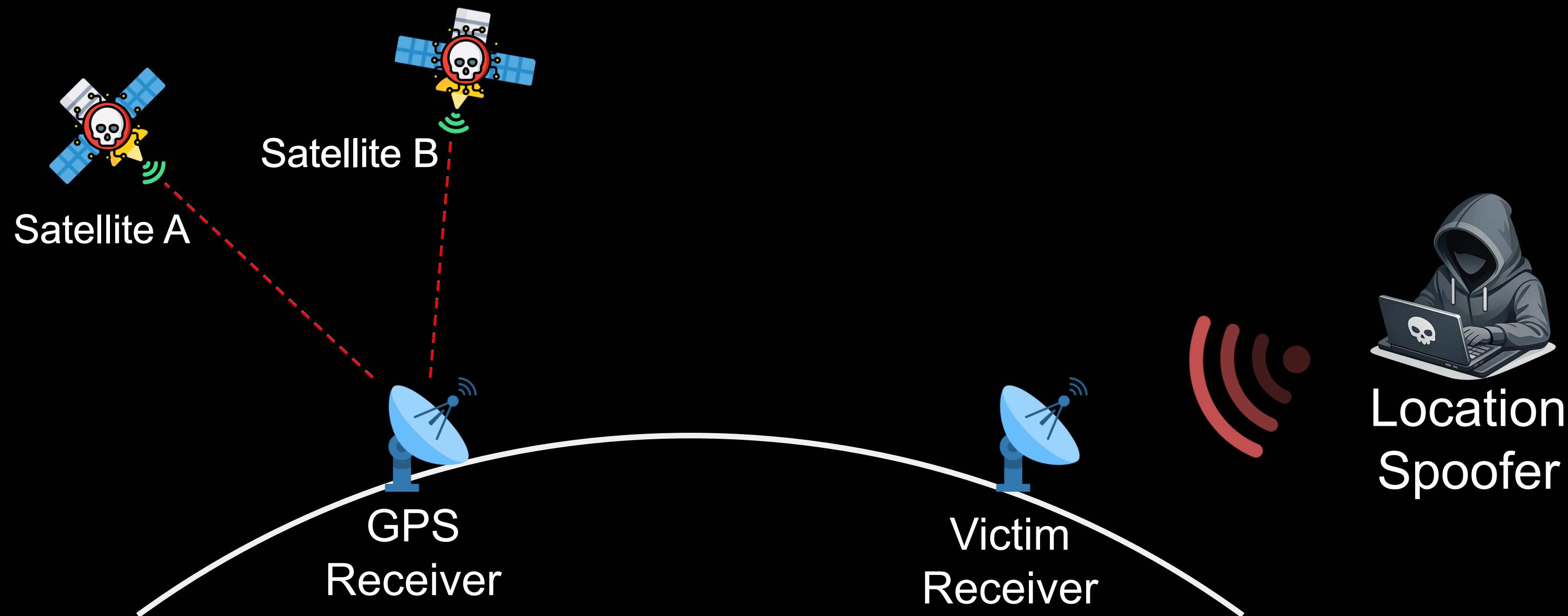


*Can an attacker still attack the system, controlling allocated frequency & power (to over/under allocate), even **without physical access to the AP?***

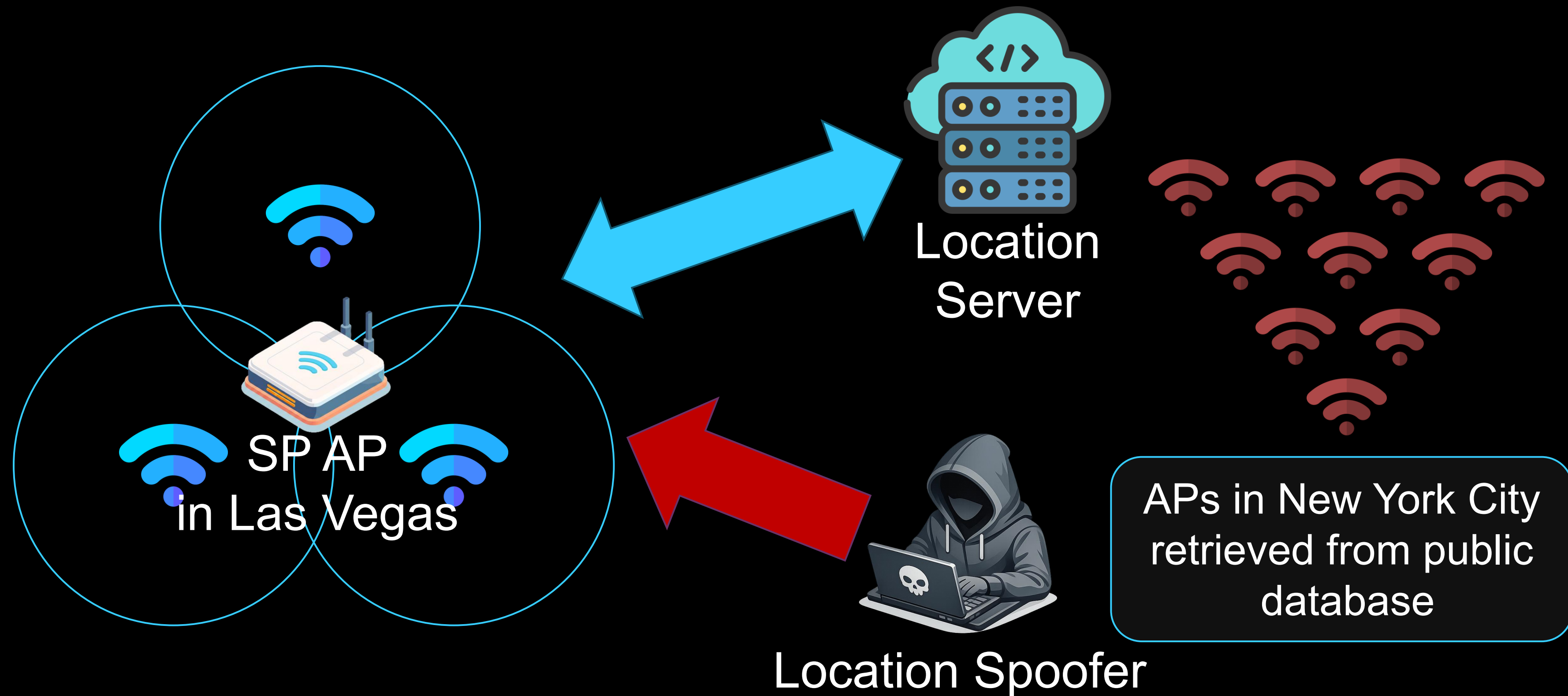
Location Spoofing Attacks against AFC Systems



Overview of GPS Spoofing



Wi-Fi Positioning Spoofing



Experiment Setup For Location Spoofing Attacks

Transmitter:

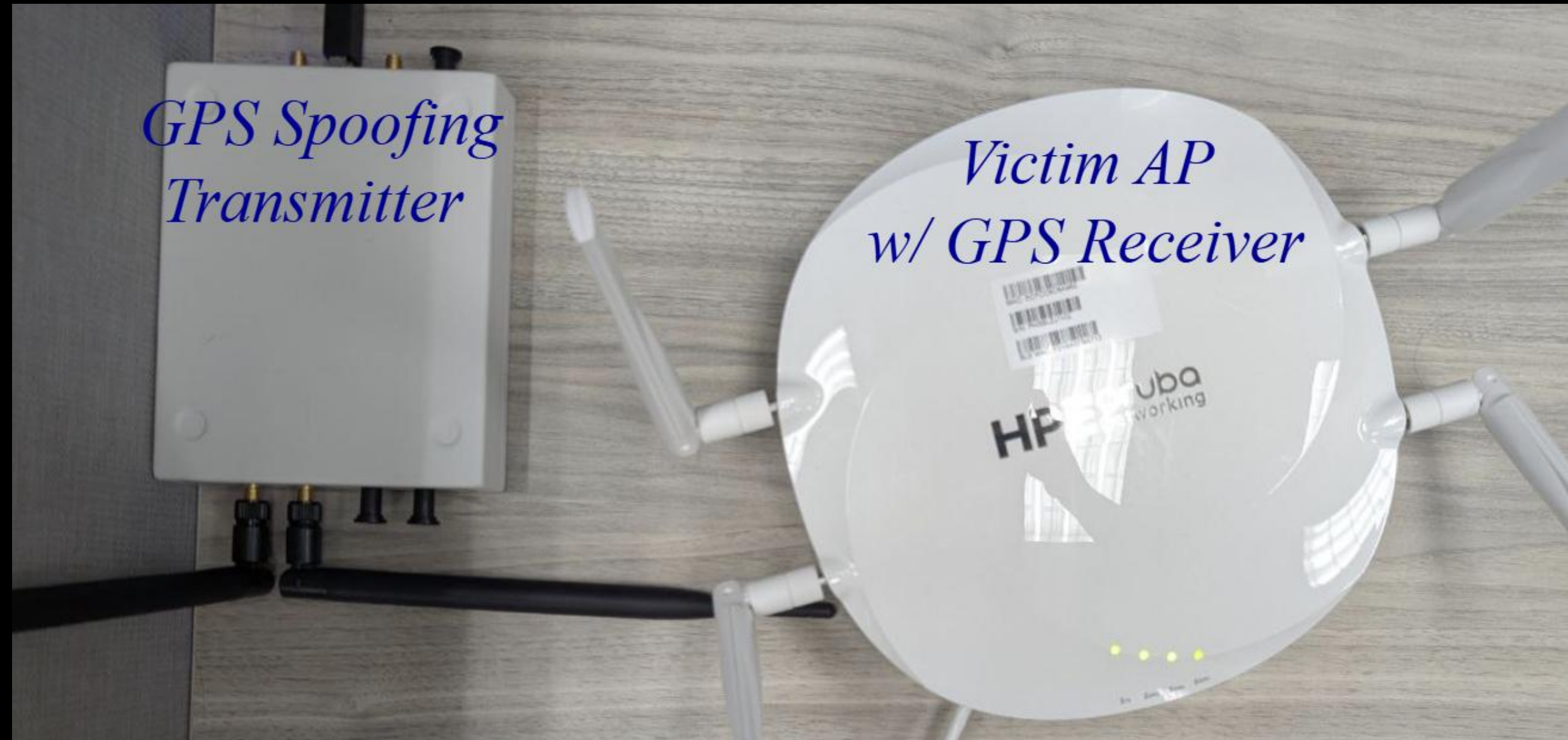
- USRP B210

Software:

- GPS-SDR-SIM (GPS)
- gr-802-11 (Wi-Fi)

APs Tested:

- HPE Aruba AP-634
- RUCKUS T670
- Ubiquiti U7 Pro Outdoor
- ROG STRIX GS-BE18000



DEMO: Wi-Fi Location Spoofing

AP Shell

```
ssh unifi_SyNSec@192.168.137.120
unifi_SyNSec@192.168.137.120's password:
```

```
***** NOTICE *****
***
* By logging in to, accessing, or using any Ubiquiti product, you are
*
* signifying that you have read our Terms of Service (ToS) and End User
*
* License Agreement (EULA), understand their terms, and agree to be
*
* fully bound to them. The use of SSH (Secure Shell) can potentially
*
* harm Ubiquiti devices and result in lost access to them and their data
*
* By proceeding, you acknowledge that the use of SSH to modify device(s)
*
* outside of their normal operational scope, or in any manner
*
* inconsistent with the ToS or EULA, will permanently and irrevocably
*
* void any applicable warranty.
*
*****
***
```



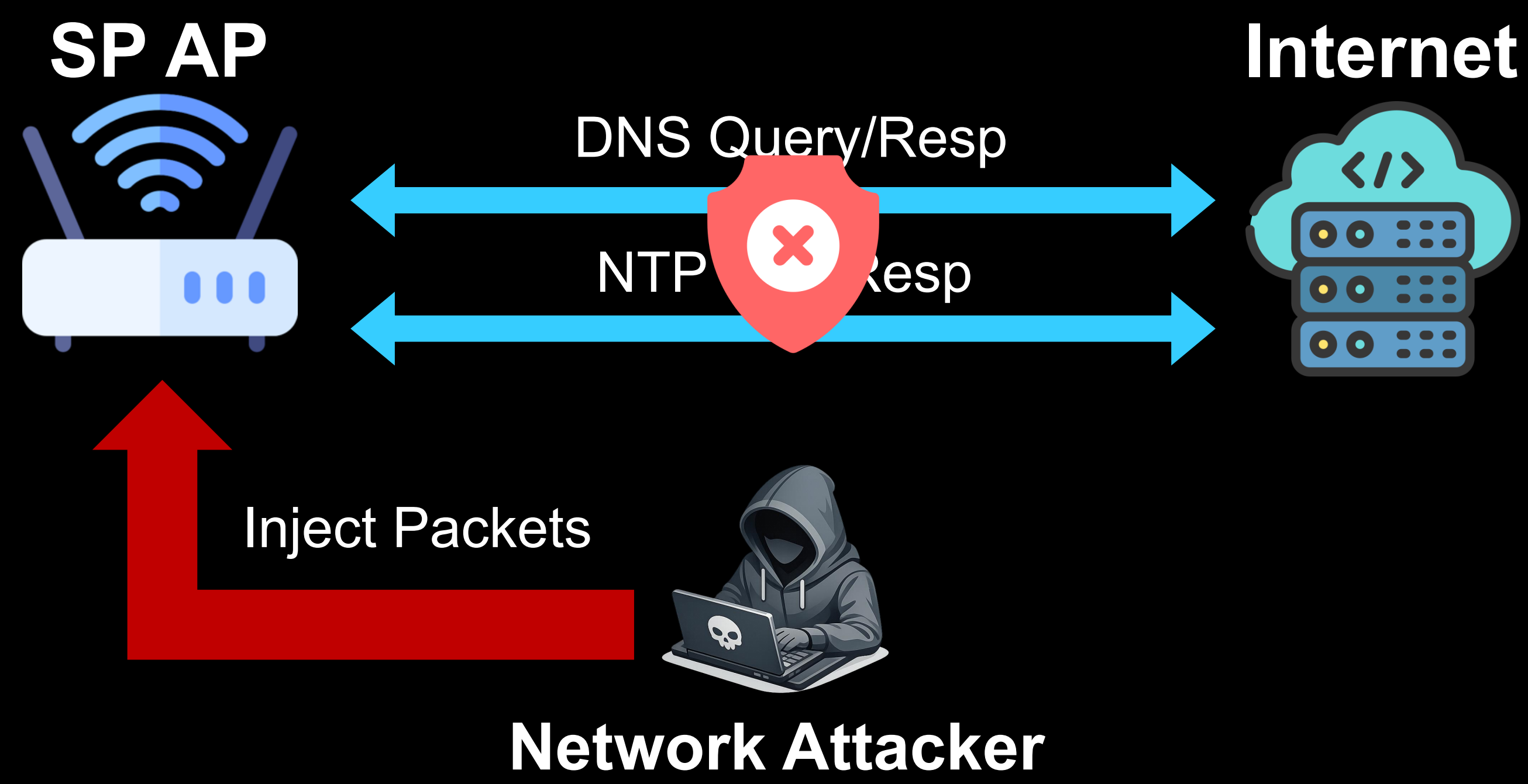
```
PRODUCT: U7-Pro-Outdoor
MAC: [REDACTED]
VERSION: 8.4.6+18068.260111.0707
```

```
Ubiquiti Inc. (c) 2010-2026      https://www.ui.com
```

```
unifi_SyNSec@U7ProOutdoor:~# qtiwifilocation
```

Spoofers Shell

DNS and NTP Attacks



Attack Summary

	A1 INTERFERE	A2 FOREIGN	A3 BAD TIME	A4 NTP	A5 DNS	A6 FORCE CVE
HPE ARUBA AP-634 GPS · Federated Wireless	●	●	●	●	●	●
RUCKUS T670 GPS · CommScope	●	●	●	○	●	○
UBIQUITI U7 PRO OUTDOOR Wi-Fi · Qualcomm	●	●	●	●	●	○
ROG STRIX GS-BE18000 Wi-Fi · Wi-Fi Alliance	●	●	●	●	●	○

●

 ATTACK PRESENTS

○

 NOT SUCCESSFUL

Interference Attack (A1)

- Generate the spoofing signal to a rural area
- The AP send the request with the spoofed coordinate to the AFC server
- The server reply with **all channels available with maximum allowed power**

Max EIRP of AFC channel																					
20MHz channel	1	5	9	13	17	21	25	29	33	37	41	45	49	53	57	61	65	69	73	77	81
Max Eirp	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0
20MHz channel	85	89	93	117	121	125	129	133	137	141	145	149	153	157	161	165	169	173	177	181	
Max Eirp	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0
40MHz channel	1	9	17	25	33	41	49	57	65	73	81	89	121	129	137	145	153	161	169	177	
Max Eirp	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0
80MHz channel	1	17	33	49	65	81	129	145	161												
Max Eirp	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0	36.0											
160MHz channel	1	33	65	129																	
Max Eirp	36.0	36.0	36.0	36.0																	
320MHz_1 channel	1																				
Max Eirp	36.0																				
320MHz_1 channel	33																				
Max Eirp	36.0																				

Max EIRP of AFC channel		
20MHz channel	1	5
Max Eirp	36.0	36.0

Denial-of-Service Attacks (A2-A5)

Impact: No allowed channels in response, the AP cannot transmit in 6 GHz bands.

A2: Spoof AP to an invalid location, e.g., a foreign country

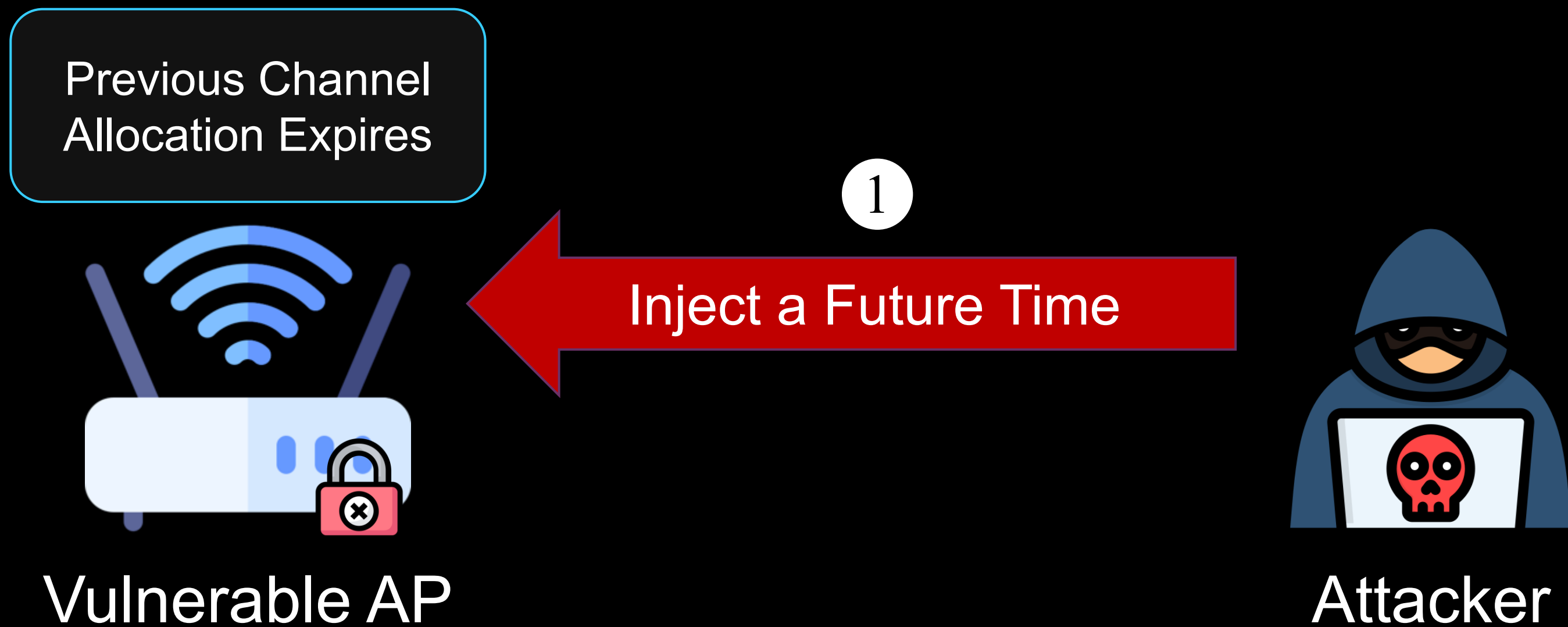
A3: Send the GPS signal using invalid time

A4, A5: DoS by NTP and DNS Spoofing

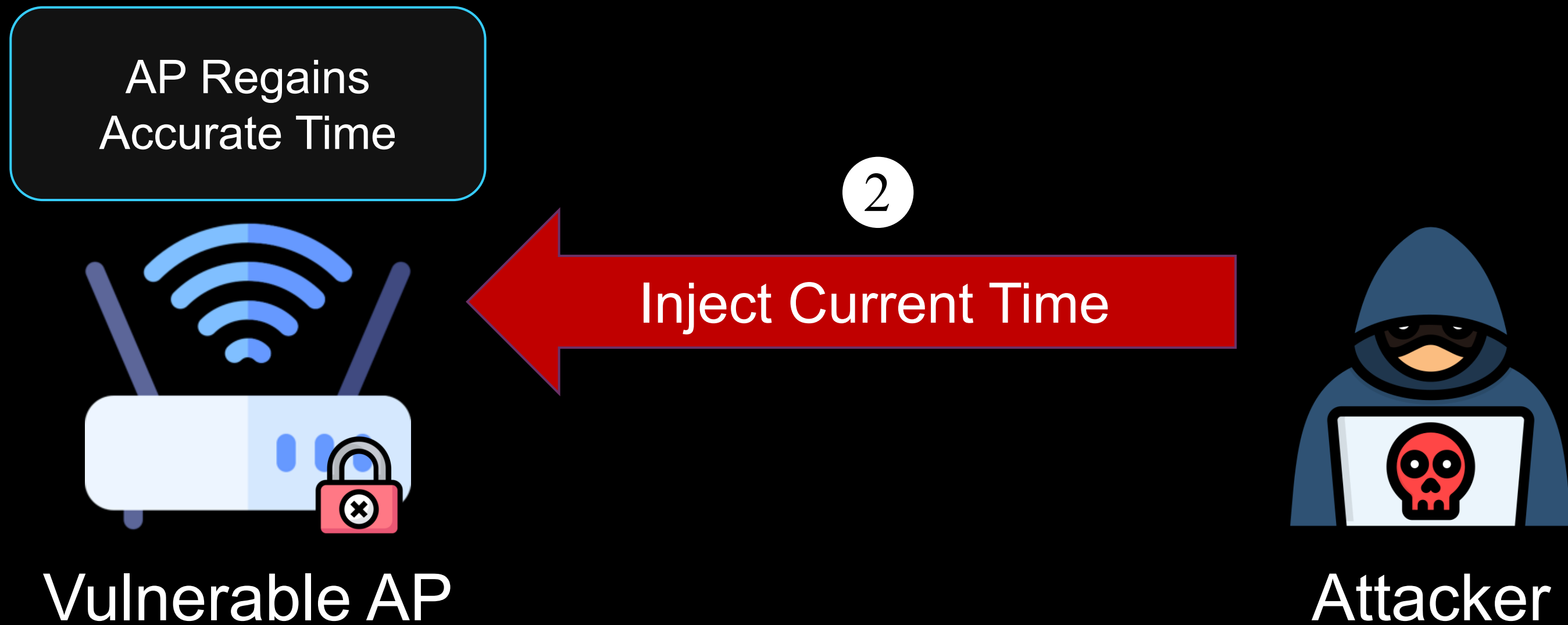
Received afc channels	

PHY Type	Allowed Channels
-----	-----
6GHz	None
6GHz 40MHz	None
6GHz 80MHz	None
6GHz 160MHz	None
6GHz 80+80MHz	None
6GHz 320MHz_1	None
6GHz 320MHz_2	None
Present time	2025-06-20 11:36:04
Expiry time	None
Country code	None
AFC channel expired	Yes
AFC channel required	Yes

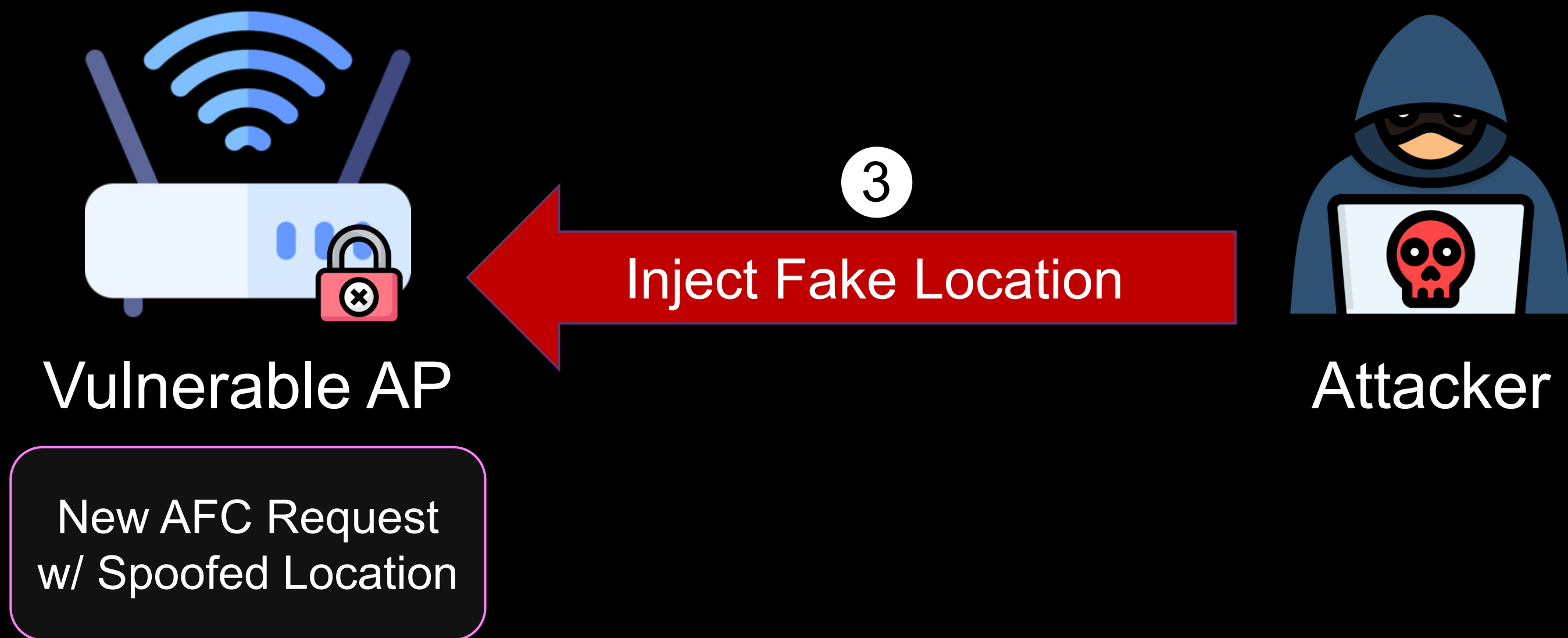
Force Location Update Attack (A6)



Force Location Update Attack (A6)



Force Location Update Attack (A6)



**WHAT IF THE ATTACKER
OWNS THE DEVICE?**

Consumer-Level APs Give User Sufficient Control

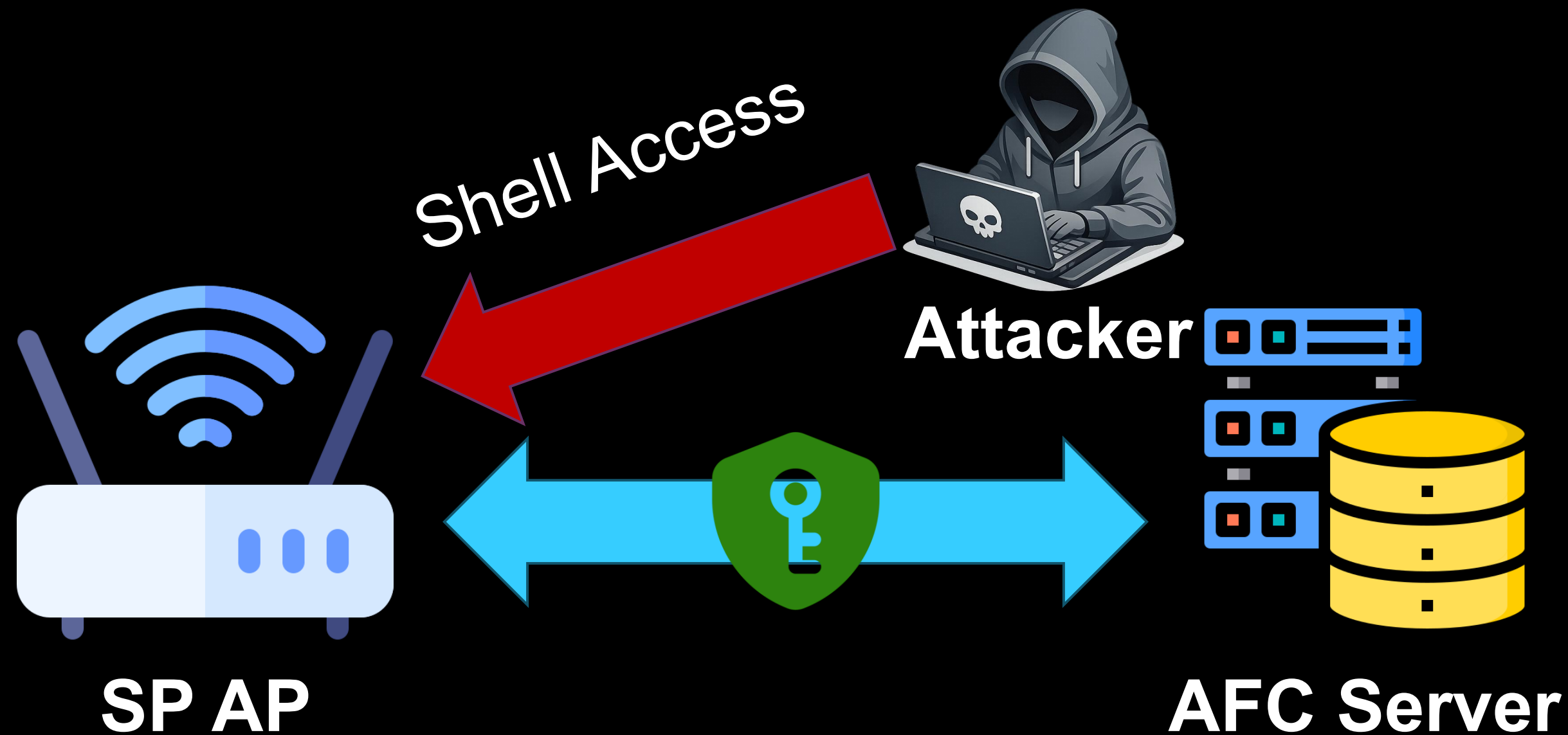


Ubiquiti U7 Pro Outdoor



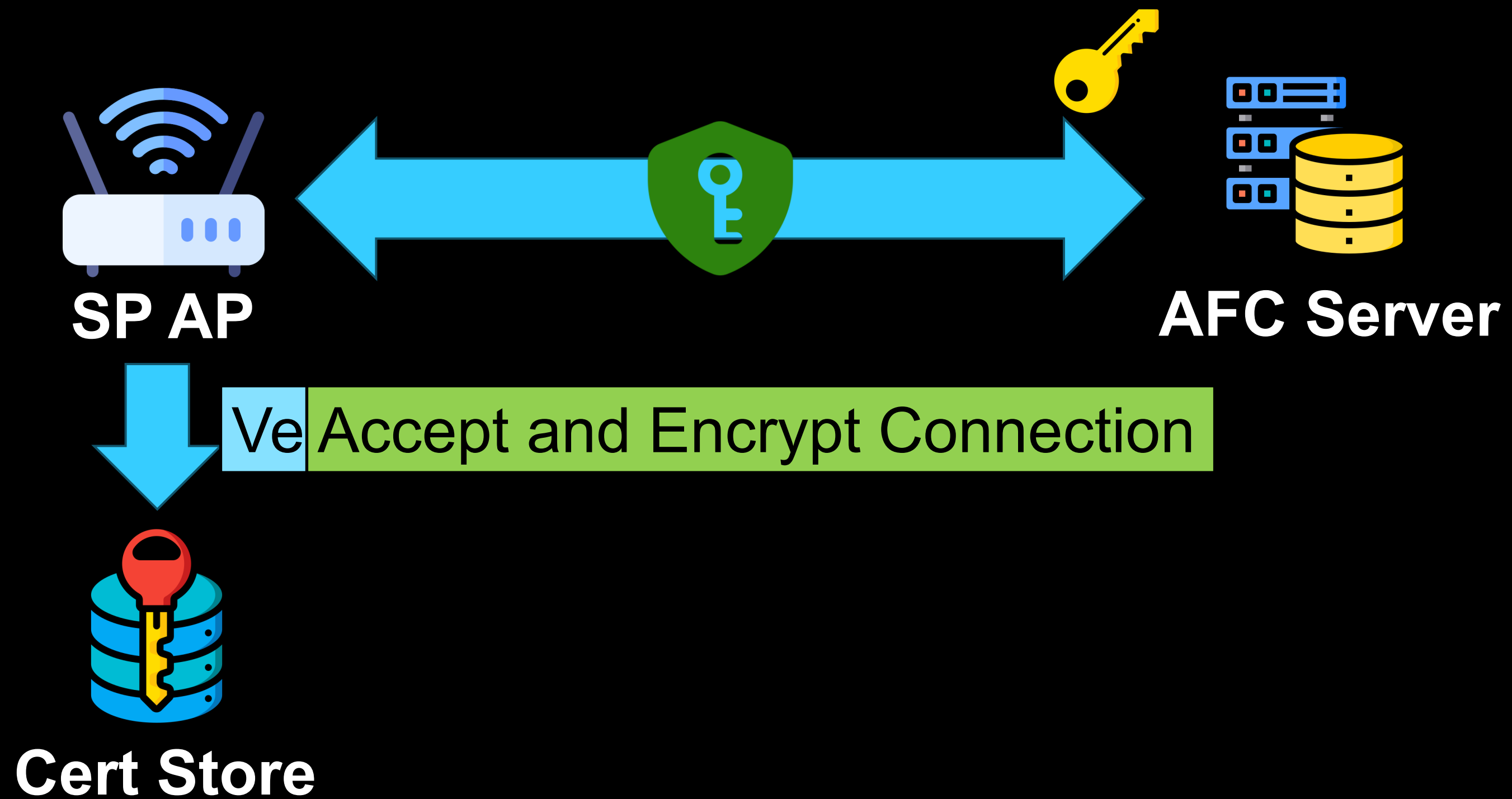
ROG STRIX GS-BE18000

Threat Model 2

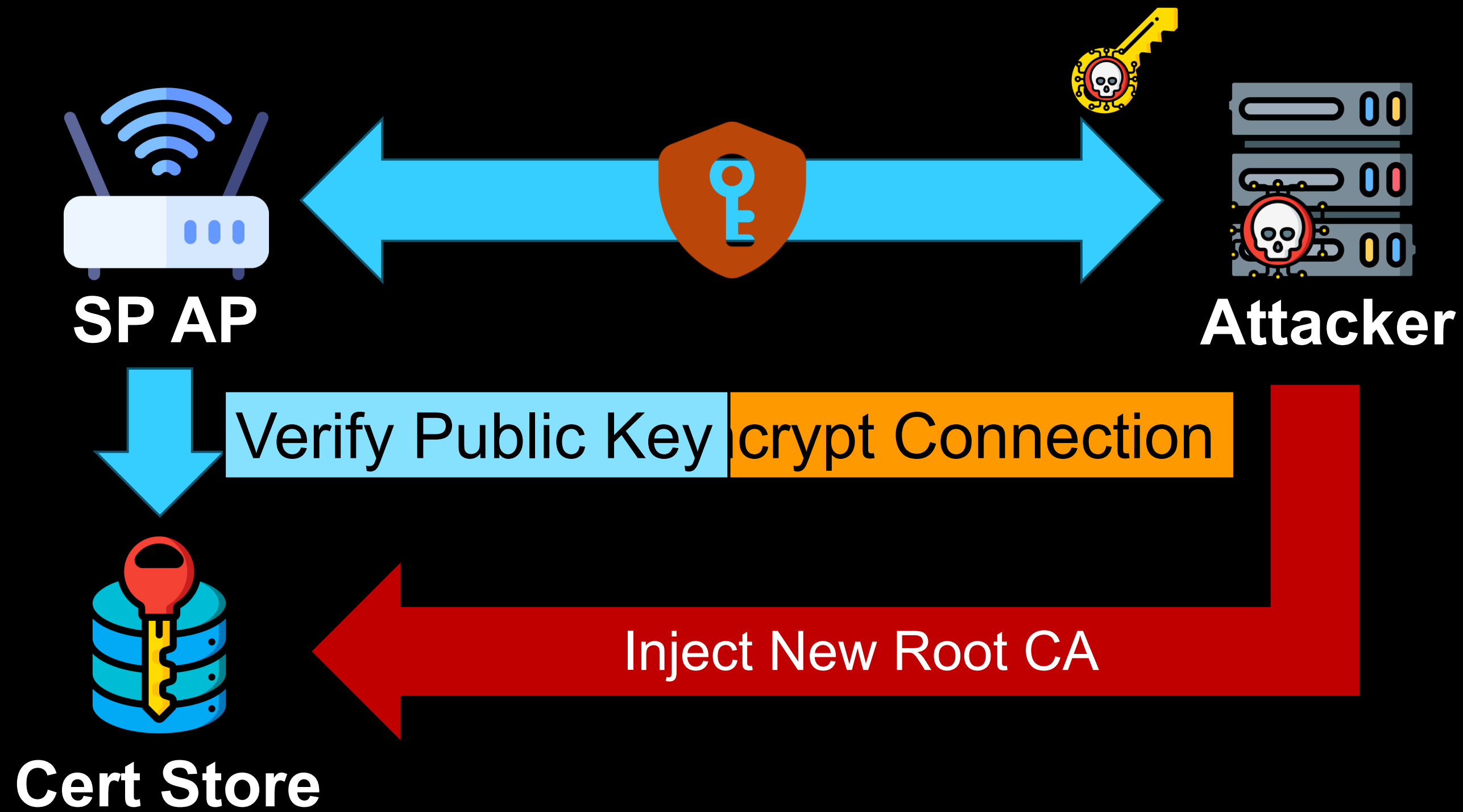


With the **user-level access** to the AP, can an attacker attack the AFC system to **manipulate the frequency and power level** transmitted from the AP?

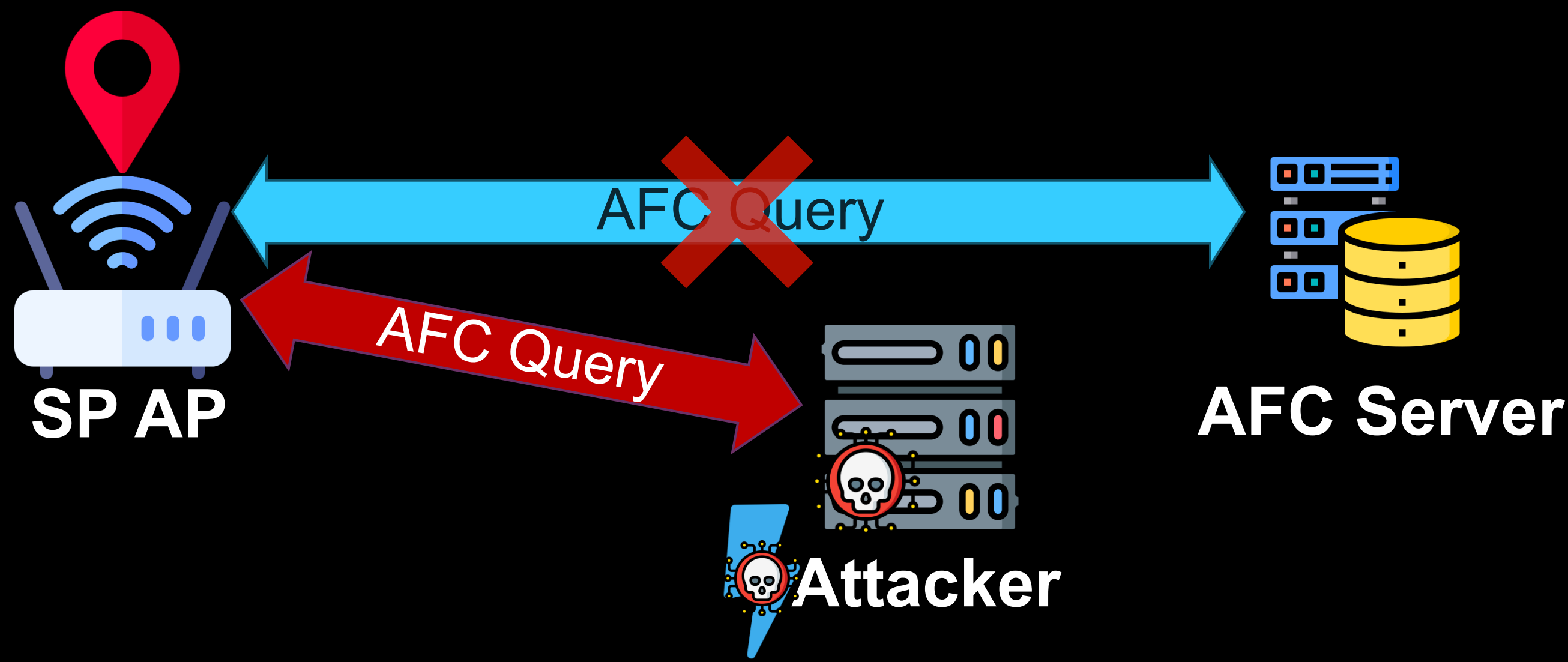
The Root of Trust



The Root of Trust (Compromised)



Forge AFC Response



Force Channel Selection Attack

Only 20 MHz available



Only 1 channel available



```
{
  "version": "1.4",
  "availableSpectrumInquiryResponses": [{
    "requestId": "2483720664",
    "response": {
      "responseCode": 0,
      "shortDescription": "Success"
    },
    "rulesetId": "US 47 CFR PART 15 SUBPART E",
    "availableFrequencyInfo": [{
      "frequencyRange": {
        "lowFrequency": 6845,
        "highFrequency": 6865
      },
      "maxPsd": 23
    }],
    "availableChannelInfo": [{
      "globalOperatingClass": 131,
      "channelCfi": [181],
      "maxEirp": [36.0]
    }],
    "availabilityExpireTime": "2025-10-21T19:08:55Z"
  }]
}
```

DEMO: Force Channel Selection Attack

AP Shell

```
ssh unifi_SyNSec@192.168.137.101
***** NOTICE *****
* By logging in to, accessing, or using any Ubiquiti product, you are *
* signifying that you have read our Terms of Service (ToS) and End User *
* License Agreement (EULA), understand their terms, and agree to be *
* fully bound to them. The use of SSH (Secure Shell) can potentially *
* harm Ubiquiti devices and result in lost access to them and their data. *
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* void any applicable warranty. *
*****
```



```
PRODUCT: U7-Pro-Outdoor
MAC: 
VERSION: 8.4.6+18068.260111.0707
```

```
Ubiquiti Inc. (c) 2010-2026 https://www.ui.com
```

```
unifi_SyNSec@U7ProOutdoor:~#
```

Attacker Shell

Force Channel Selection Attack



U7 Pro Outdoor

Connected To -

 AirView

TX Retries

Low (0%)

11:10 PM

11:10 AM

Now

↓ 512 bps

↑ 3.13 Kbps

9m 51s

Ch. 11 (2.4 GHz, 20 MHz)

2x2

WiFi 7

 0

Ch. 48 (5 GHz, 40 MHz)

2x2

WiFi 7

 0

Ch. 181 (6 GHz, 20 MHz)

2x2

WiFi 7

 0

Ch. 181 (6 GHz, 20 MHz)

2x2

WiFi 7

 0

AP forced to use channel 181



U7 Pro Outdoor

Connected To -

 AirView

 6 GHz is currently disabled. The AP did not receive an AFC response for this location and will automatically retry. [Learn more](#)



TX Retries

Low (0%)

11:05 PM

11:05 AM

Now

↓ 1.22 Kbps

↑ 38.8 Kbps

3m 35s

Ch. 11 (2.4 GHz, 20 MHz)

2x2

WiFi 7

 0

Ch. 48 (5 GHz, 40 MHz)

2x2

WiFi 7

 0

Ch. Auto (6 GHz, 20 MHz)

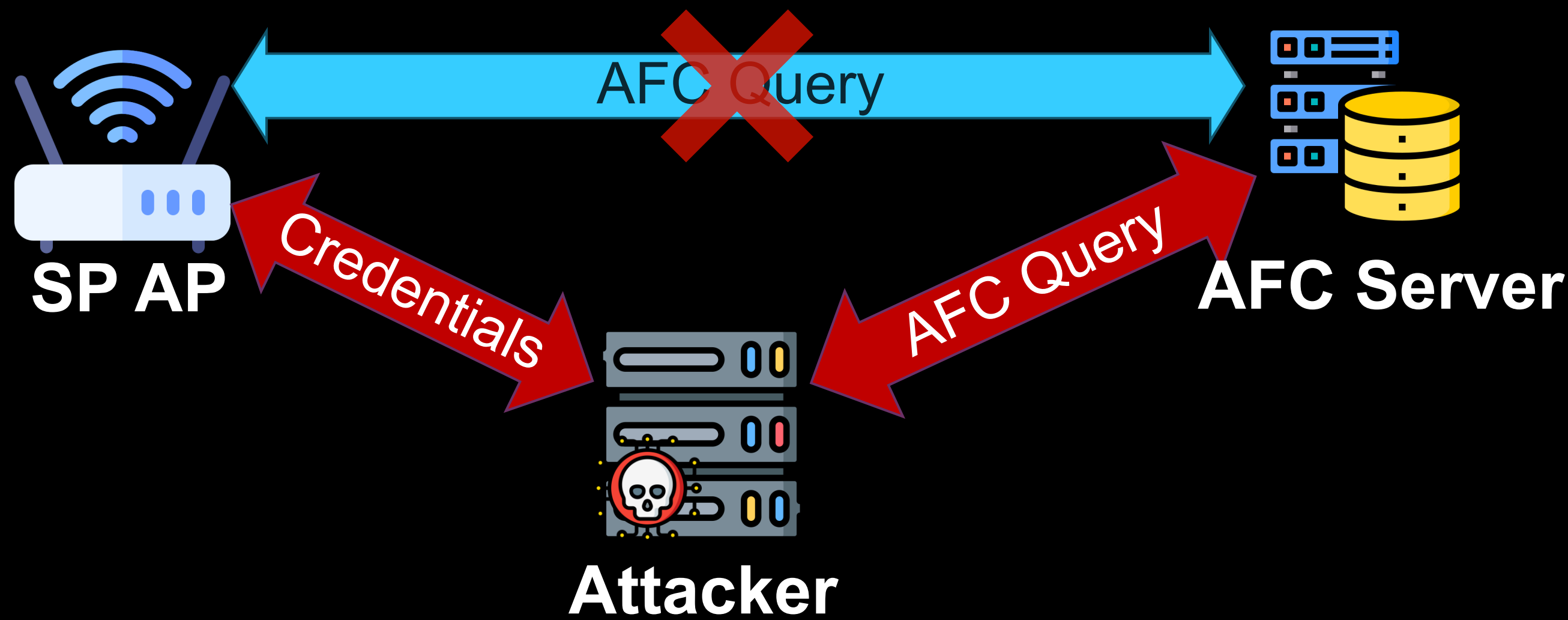
2x2

WiFi 7

 0

6 GHz channel disabled

Send Requests From Client to Server



The Ubiquiti/Qualcomm Approach: Shared Secret+JWT Token

1. Find the HTTP endpoint for AFC
2. Extract the shared secret from the connection
3. Use the shared secret to obtain JWT token
4. Attach the JWT token to the AFC request

TLS MitM · U7-PRO-OUTDOOR → QUALCOMM

```
POST /device/0c18f25c-XXXX-XXXX-XXXX-XXXXXXXXXXXX/ api.qcs
{"api_version":1,"request_type":"request_authToken", ← the tuple
 "uid":"16165XXXX","oem_id":"536","prd_id":"42662",
 "shared_secret":"HxnT5bXXXXXXXXXXXXXXXXXXXX", ← cleartext in the session
 "application":"AFC"}
```

```
HTTP/1.1 200 OK Server: gunicorn
{"status":"OK","auth_token":"eyJhbGciOiJIUzI1NiIs..."} 
```

```
JWT header {"alg":"HS256","typ":"JWT"}
claims {"exp":1767293788, 30-day life
 "app":"AFC",
 "uid":"7881175f-XXXX-XXXX-XXXXXXXXXXXX",
 "certs":[["CA","6545A-U7PROO"],
 ["US","SWX-U7PROO"]]}
```

```
POST /afc_client_api/afc_api/ afcapi.qcs
Authorization:eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9... ← no Bearer
```

Serial, shared secret, service path and token identity are masked.

The ASUS/Wi-Fi Alliance Approach: Mutual TLS

1. Find the HTTP endpoint for AFC
2. Extract client certificate from the device
3. Use the certificate to connect to the server

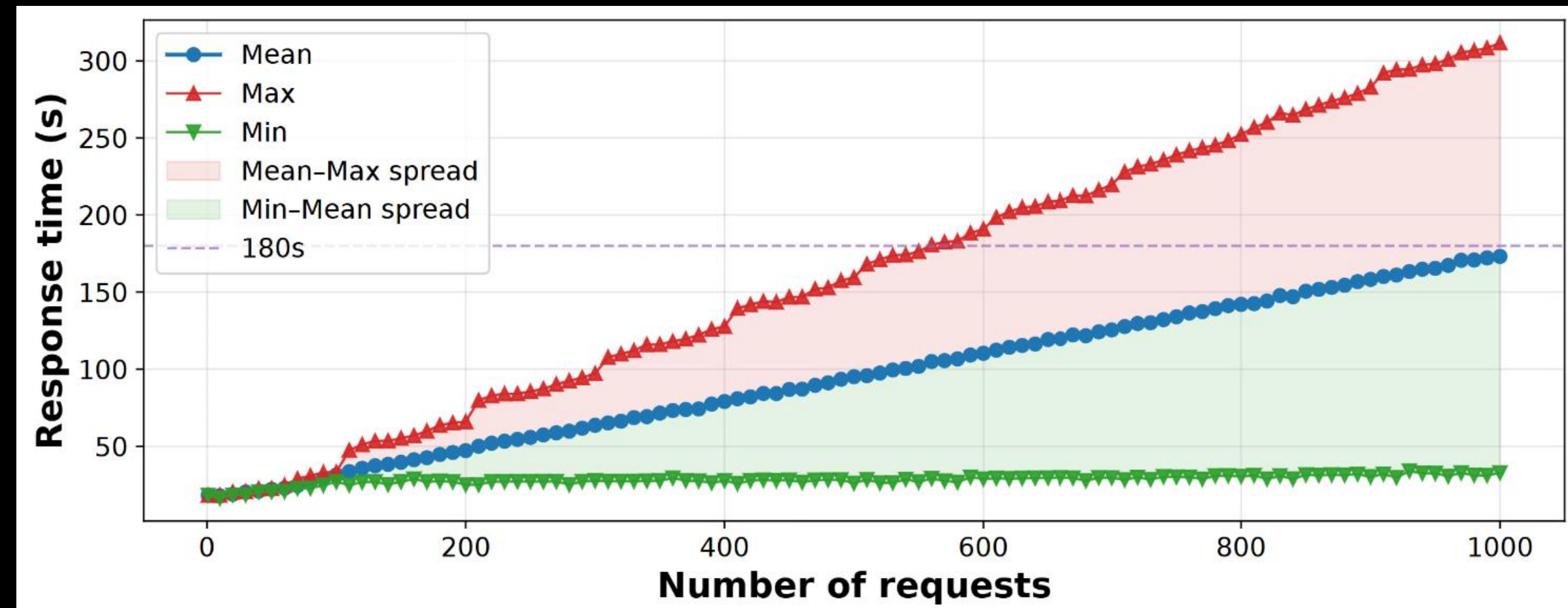
```
ASUS GS-BE18000 · SHELL
```

```
admin@GS-BE18000-CB50:/# nvram show | grep afc_  
afc_cert_id=MSQ-XXXXXXX  
afc_dev_serial_no=XXXXXXXXXXXXXXXXX  
afc_freq_range=5925,6425;6525,6875  
afc_geo_from=source_wifi  
afc_in_wifiaplist={"considerIp":"false","wifiAccessPoints":[  
  {"macAddress":"XX:XX:XX:XX:XX:XX","signalStrength":-20}, ...  
afc_insecure=0  
afc_loc_height_type=AGL  
afc_mtls_cert=/jffs/.sys/afc/afc_mtls_cacert.pem ← cert path  
afc_op_class=131,132,133,134,137  
afc_out_lat=XX.XXXXXXX  
afc_out_lng=-XX.XXXXXXX  
afc_reg_rules=US_47_CFR_PART_15_SUBPART_E  
afc_sp=1  
afc_url=https://mtls-access.afc.wi-fi.com/api/XXXXXXXXX ← endpoint
```

Serial number, MAC addresses, coordinates and the API token are masked.

Resource Exhaustion on AFC Server

- AFC request consumes a considerable amount of computation power of the server
- A single request can take 10-20 seconds with our testbed and Open AFC
- Attacker can launch DDoS attacks with the capability to send the requests as a device

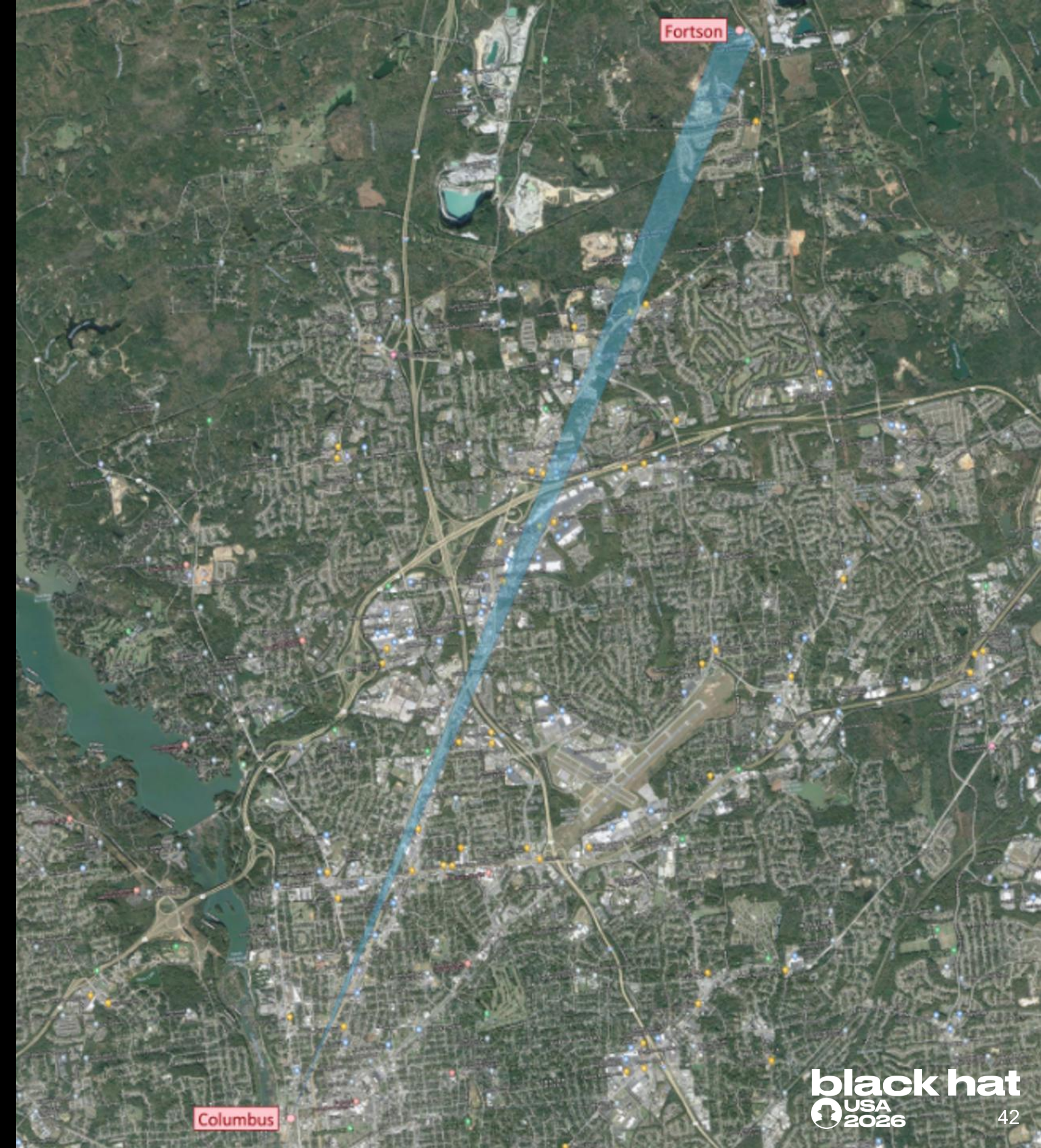


Concurrent Requests VS. Response Time on Open AFC

How are Interference Impacting FS?

- Columbus and Fortson, GA - 9.5 mi
- inside shops beneath a working utility 6 GHz microwave link
- low-power, indoor, AFC-exempt AP were enough to cause harmful interference
- beacon frames alone (no connected client, no data): ~32 dB past the threshold.

PEAK INTERFERENCE (I/N) INTO THE FS RECEIVER VS. DISTANCE



How an Attacker May Exploit AFC?

RECON FS Record

FS link information is public

Miami-Dade 6 GHz Microwave Call Signs

- WI)
- WC 21
- WC 23
- WC 69
- WC 79
- WC 86
- WC 88
- WC 92
- WC 95
- WC 96
- WC 99
- WC 03
- WC 05
- WC 08
- WC 14
- WC 17
- WI 27
- WI 38
- WI 39

 **Federal Communications Commission**

Universal Licensing System

FCC > WTB > ULS > Online Systems > License Search

License Search

Search Results

 [New Search](#)  [Refine Search](#)  [Printable Page](#)  [Query Download](#)  [Map Licenses](#)

Specified Search

Radio Service = **CF, MG, MW**

Name like **MIAMI-DADE**

Status = **Active**

Frequency Upper Band >= **5925**

Frequency Assigned <= **7125**

How an Attacker May Exploit AFC?

RECON FS Record

FS link information is public

Miami-Dade 6 GHz Microwave Call Signs

- W 0
- W 21
- W 23
- W 69
- W 79
- W 86
- W 88
- W 92
- W 95
- W 96
- W 99
- W 03
- W 05
- W 08
- W 14
- W 17
- W 27
- W 38
- W 39

Federal Communications Commission

Universal Licensing System

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FCC Site Map

License Search

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Printable Page

Query Download

Map Licenses

Specified Search

Radio Service = CF, MG, MW
Name like MIAMI-DADE
Status = Active
Frequency Upper Band >= 5925
Frequency Assigned <= 7125

Matches 1- 10 (of 27)

Page 1 2 3

NEXT ▶

	Call Sign/Lease ID	Name	FRN	Radio Service	Status	Expiration Date
1	WP 0	MIAMI-DADE COUNTY	0001802735	MW	Active	10/17/2031
2	WQ 21	Miami-Dade County	0001802735	MW	Active	03/19/2029
3	WQ 23	Miami-Dade County	0001802735	MW	Active	03/19/2029
4	WQ 10	MIAMI-DADE COUNTY	0001802735	MW	Active	01/26/2030
5	WQ 69	MIAMI-DADE COUNTY	0015402795	MW	Active	09/30/2030
6	WQ 79	MIAMI-DADE COUNTY	0015402795	MW	Active	09/30/2030
7	WQ 86	MIAMI-DADE COUNTY	0015402795	MW	Active	09/30/2030
8	WQ 88	MIAMI-DADE COUNTY	0015402795	MW	Active	09/30/2030
9	WQ 92	MIAMI-DADE COUNTY	0015402795	MW	Active	09/30/2030
10	WQ 95	MIAMI-DADE COUNTY	0015402795	MW	Active	09/30/2030

Page 1 2 3

NEXT ▶

PA= Pending Application(s)
TP= Termination Pending
L= Lease

How an Attacker May Exploit AFC?

RECON FS Record

FS link information is public

ULS License

Microwave Public Safety Pool License - WF[REDACTED] - MIAMI-DADE COUNTY

[New Search](#) [Refine Search](#) [Return to Results](#) [Printable Page](#) [Reference Copy](#) [Map License](#)

MAINADMINLOCATIONSPATHSMAP

Call Sign	WF[REDACTED]0	Radio Service	MW - Microwave Public Safety Pool
Status	Active	Auth Type	Regular
Dates			
Grant	09/24/2021	Expiration	10/17/2031
Effective	12/29/2022	Cancellation	
Control Points			
None			
Licensee			
FRN	00[REDACTED]35 (View Ownership Filing)	Type	Governmental Entity
Licensee			
MIAMI-DADE COUNTY 5680 SW 87th Ave MIAMI, FL 33173 ATTN ITD Radio Communications Division		P:(305)[REDACTED]909 F:(305)[REDACTED]72 E:luna[REDACTED]@miami-dade.gov	
Contact			
MIAMI-DADE COUNTY Miguel D Luna 5680 SW 87th Ave MIAMI, FL 33173 ATTN ITD Radio Division		P:(305)[REDACTED]909 F:(305)[REDACTED]96 E:luna[REDACTED]@miami-dade.gov	
Microwave Data			
Oper Type	Permanent Fixed Point to Point	Station Class	FXO - Operational Fixed
Ownership and Qualifications			
Radio Service Type	Fixed		
Regulatory Status	Private Comm	Interconnected	No

How an Attacker May Exploit AFC?

RECON FS Record

FS link information is public

MAINADMINLOCATIONSPATHSMAP

Call Sign	WPJE410	Radio Service	MW - Microwave Public Safety Pool
2 Total Locations 10 Locations per Summary Page			
Fixed Transmit Location 1:			
MIAMI [REDACTED] GARAGE #4 MIAMI, FL MIAMI-DADE County		Coordinates	25-4 [REDACTED] 5.1 W
Site Elevation (AMSL)	1.5m	Height w/o Appurtenances	42.7m
ASR #/File #	N/A	Height w/ Appurtenances	42.7m
Support Structure Type			
NEPA Required			
Quiet Zone Notification Date		Quiet Zone Consent	
Is coordination with Canada required?			
Is coordination with Mexico required?			
Special Conditions	None		
Other Locations			
Location 2 : 1049			
Type	Receive Location	Coordinates	25- [REDACTED] 48.1 W
Site Elevation (AMSL)	3.0m	Height w/o Appurtenances	
ASR #/File #	N/A	Height w/ Appurtenances	
Support Structure Type			

How an Attacker May Exploit AFC?

RECON **FS Record**

FS link information is public

MAINADMINLOCATIONSPATHSMAP

Call Sign

W 1.0

Radio Service

MW - Microwave Public Safety Pool

License Geography



How an Attacker May Exploit AFC?

RECON **FS Record**

FS link information is public

SPOOF **Side Channel**

Off-path spoofing of
location/time side channel

ATTACK **Re-query**

Wait for 24-hour AFC re-
query or force

How an Attacker May Exploit AFC?

RECON **FS Record**

FS link information is public

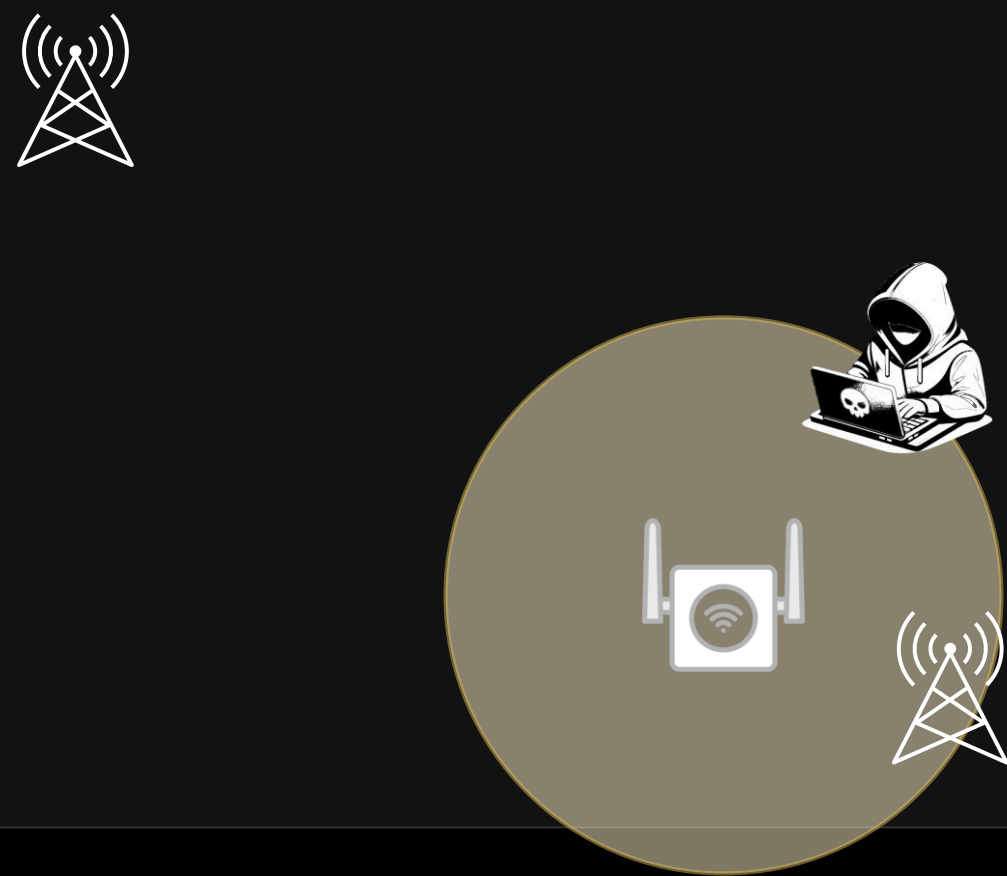
SPOOF **Side Channel**

Off-path spoofing of location/time side channel

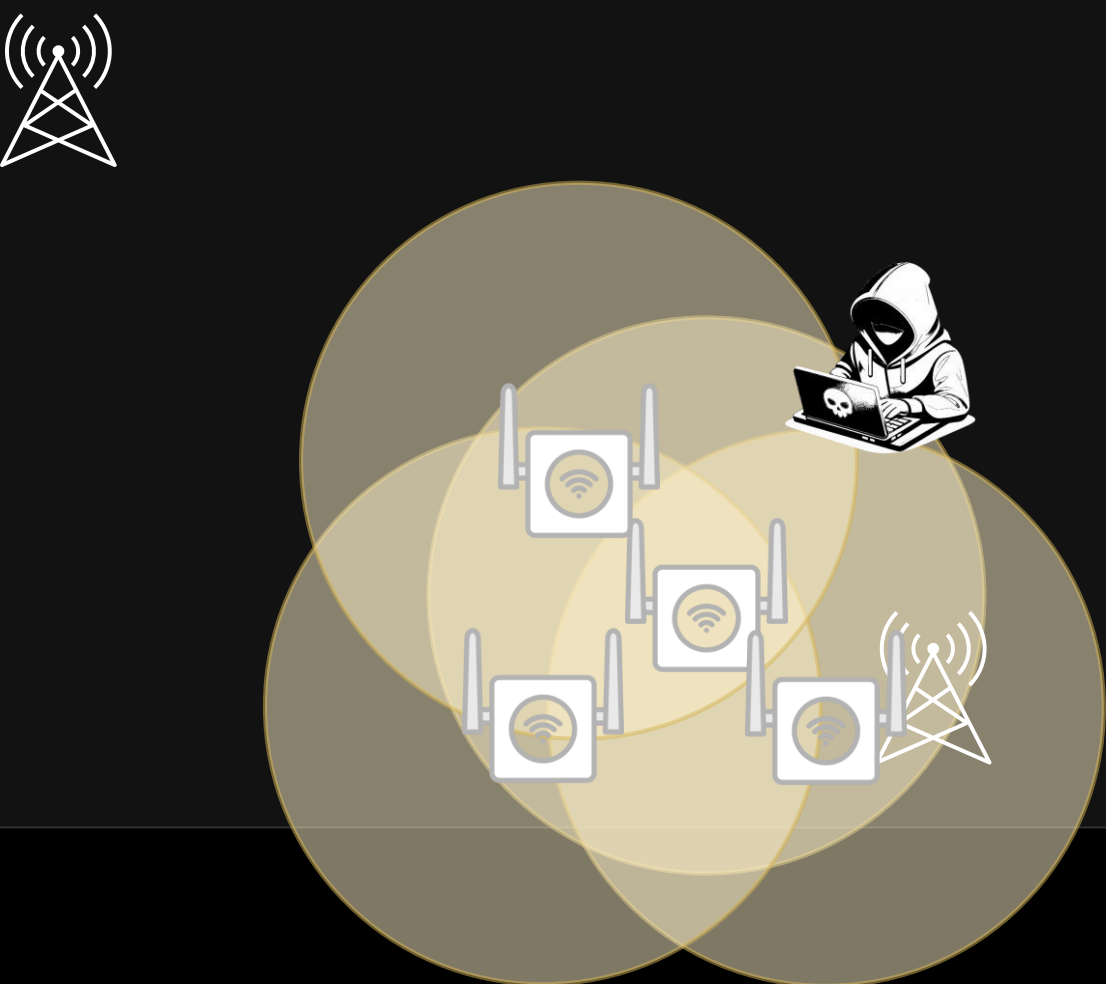
ATTACK **Re-query**

Wait for 24-hour AFC re-query or force

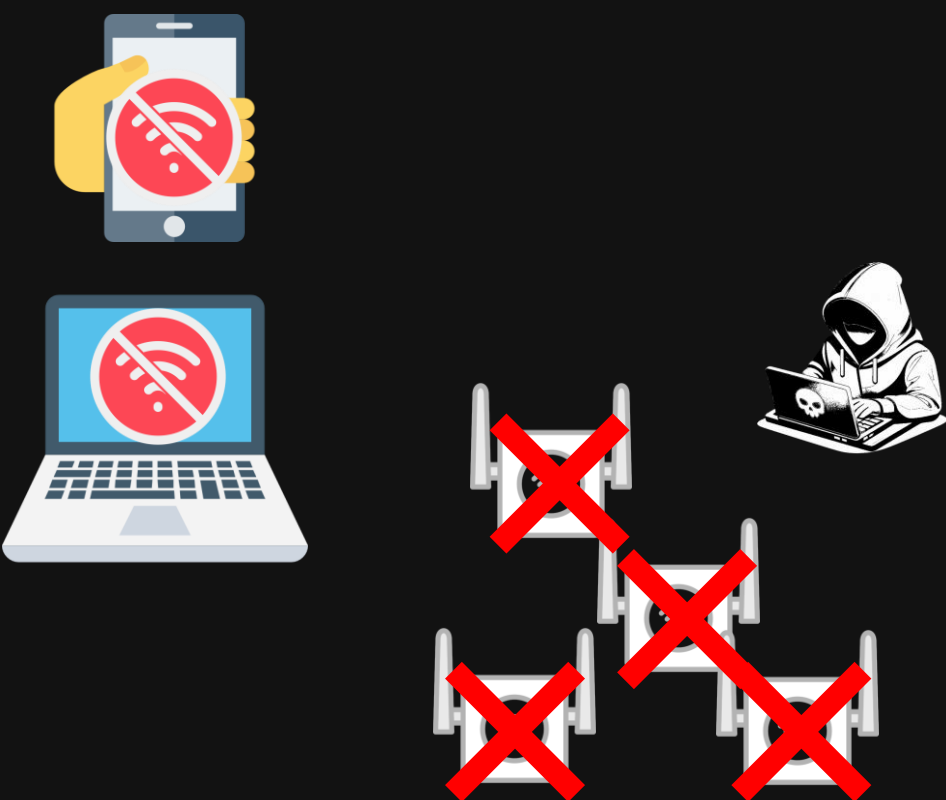
1 Interfere



2 Coordinated



3 DoS



How an Attacker May Exploit AFC?

RECON **FS Record**

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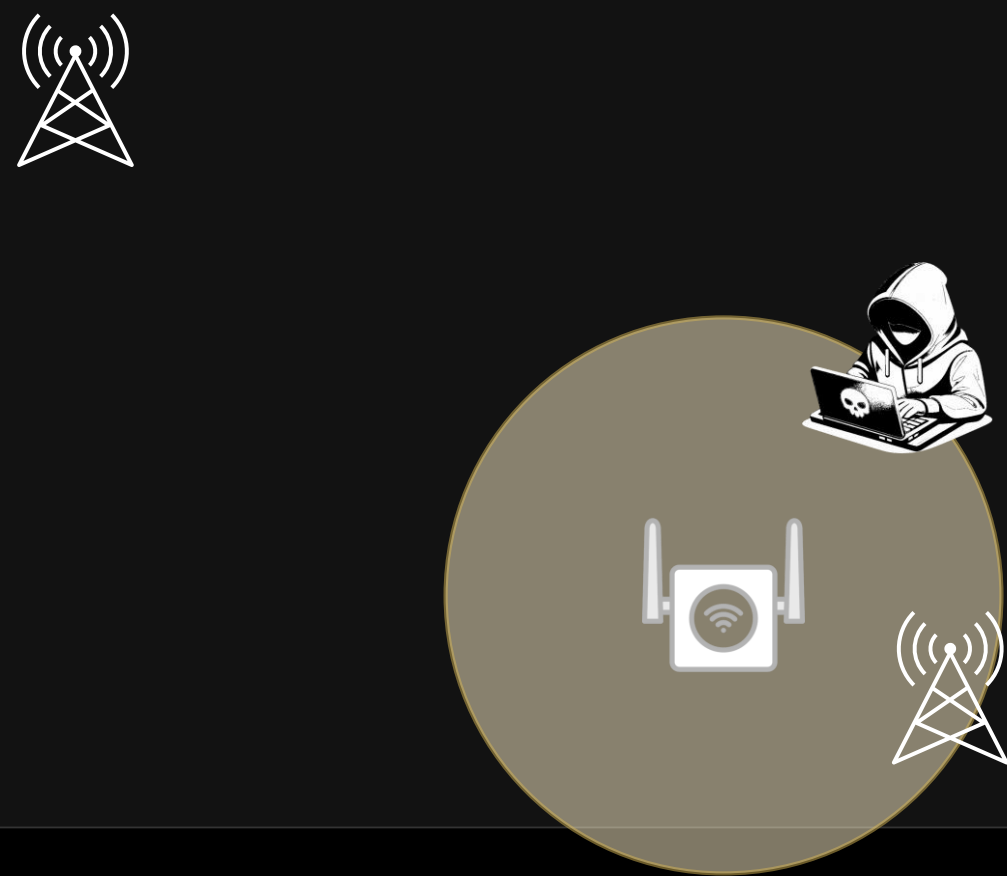
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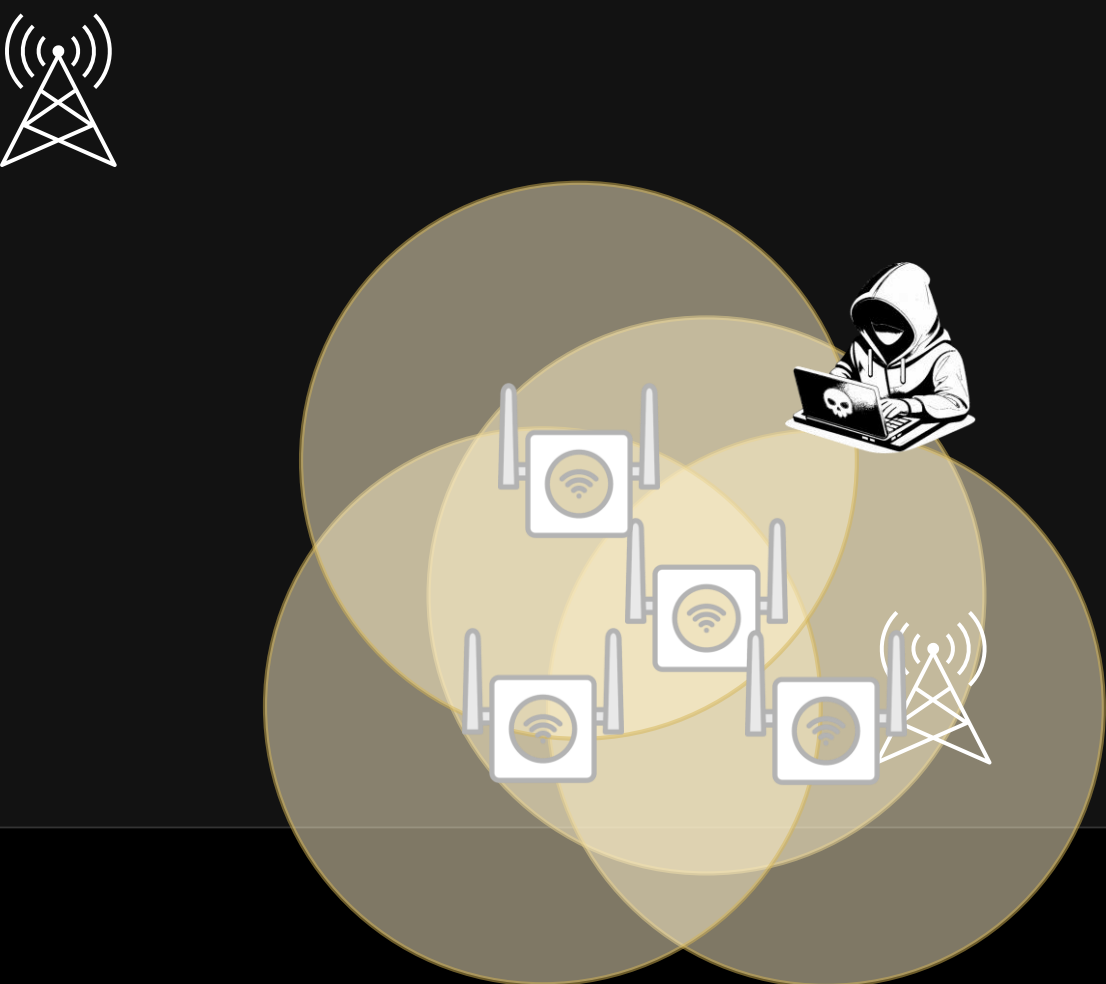
EXIT **Persist Effect**

Attacker can stop transmission while AP keeps broadcast/DoS

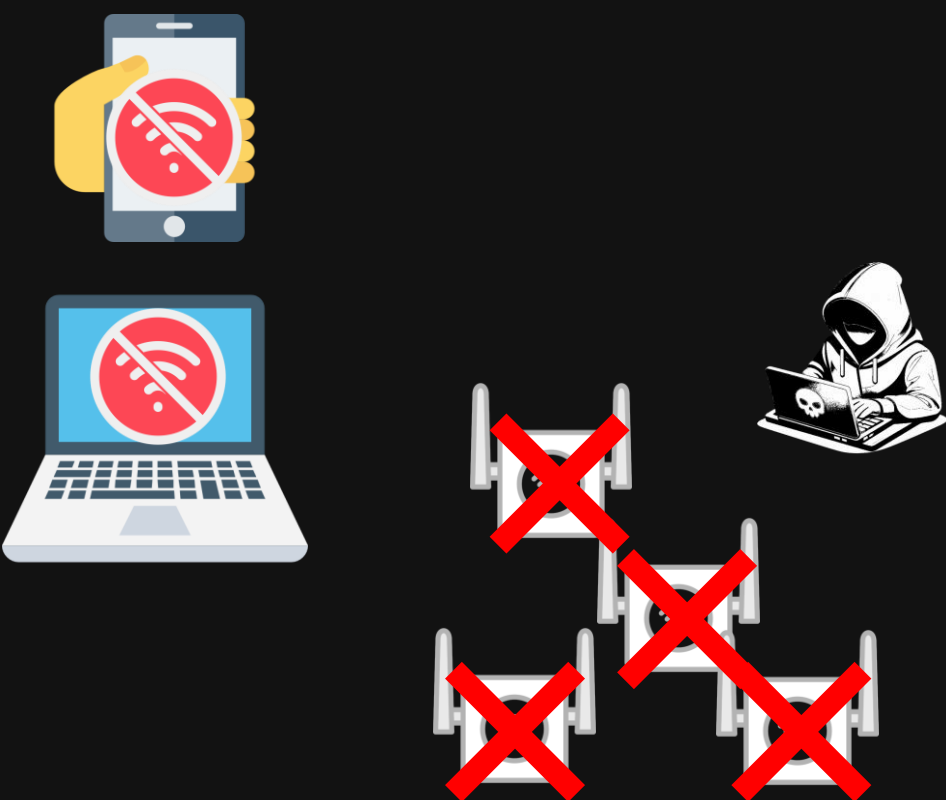
1 Interfere



2 Coordinated



3 DoS



SOLUTIONS?

Geofencing

- Standard Power APs are not designed for mobile use
- Pre-define a small possible operation area and stop transmission if outside
- Discussed and recommended by some vendors



Securing Device and Server Implementations



ON THE AP

Lock down AFC parameters

No user access to the certificate and key, or to the shared secrets.

ON THE TLS LINK

Certificate pinning

The AP rejects any root CA an attacker injects into its store.

ON THE AFC SERVER

Rate limit requests

Per device and per IP address — one query costs 10–20 s of compute.

TAKEAWAYS

Takeaways

- The security of the AFC system is critically dependent on the integrity of its inputs and implementation.
- This trust model in spec is incomplete, as low-cost spoofing attacks can cause severe interference or denial-of-service to critical incumbent systems.
- No device being tested is completely secure and can all lead to harmful interference and large-scale DoS attacks when attacked.

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